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Paleomagnetic results from a crosscutting system of northwest and north–south trending diabase dikes in the North Carolina Piedmont

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Abstract

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North–south and northwest trending suites of diabase dikes which crosscut one another have been identified in the North Carolina Piedmont from the interpretation of aeromagnetic data and field relationships (e.g., Popenoe and Zietz, 1977; Burt et al., 1978). The crosscutting relationships and areal distribution of the dikes suggest that they were emplaced during two different stress regimes representing different ages or phases of the Early Mesozoic rifting of the North American margin.

A paleomagnetic study was conducted on the two crosscutting dike suites of the North Carolina Piedmont in order to establish whether the time differential in the emplacement of the dike suites was significant or not and to confirm relative ages. Results from 11 NW trending diabase dike sites yield a VGP of 52.8° N, 60.2° E. The VGP for 15 N–S trending diabase dike sites is located at 71.5° N, 53.5° E. The Fisherian statistics of the pole position for both suites are significantly different from one another. This therefore suggests a relative age differential for the emplacement of diabase dikes with the NW suite interpreted as being relatively older than the N–S suite of dikes.

The pole positions of both suites plot off the APW path for North America. To bring these aberrant poles into near coincidence with the APW path for North America would require the tilting of the Piedmont westwards approximately 15°. However, there is no geological evidence of a measurable amount of a probably complex post Jurassic tilting of the Piedmont province of this region.

Introduction

The systematic regional distribution of diabase dikes in eastern North America was first pointed out by King (1961, 1971). Since then the dikes have been related to the tensional tectonic events associated with the breakup of Pangea in the Early Mesosoic (De Boer, 1967; May, 1971; Swanson, 1982).

Throughout eastern North America, the Mesozoic diabase has been the subject of numerous geochemical, paleomagnetic and geochronological studies. Most work on the paleomagnetism of diabase dikes, flows, and sills has been done in the northern Appalachians (e.g., Irving and Banks, 1961; Opdyke, 1961; Larochelle and Wanless, 1966; De Boer, 1967, 1968; Larochelle, 1967, 1971; Carmichael and Palmer, 1968; Beck, 1972; Smith and Noltimier, 1979; Hodych and Hayatsu, 1980; Sequin et al., 1981). Only a few studies on diabase dikes (De Boer, 1968; Watts, 1975; Bell et al., 1980; Dooley and Smith, 1982) and subsurface flows (Phillips, 1983) have been carried out in the southern Appalachians.

There are numerous geochronological studies of diabase in the Appalachians (e.g., Erickson and Kulp, 1961; Larochelle and Wanless, 1966; Carmichael and Palmer, 1968; De Boer, 1968;

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Armstrong and Besancon, 1970; Dallmeyer, 1975; Deininger et al., 1975; Dooley and Wampler, 1977, 1983; Hayatsu, 1979; Sutter and Smith, 1979; Dooley and Smith, 1982; Lanphere, 1983).

Two trends of diabase dikes (a northwest and north–south trending suite) have been identified in the North Carolina Piedmont on the basis of interpretation of various compilations of aeromagnetic and field data (e.g., Popenoe and Zietz, 1977; Burt et al., 1978). The NW trending dikes are found throughout the southern Appalachian Piedmont extending from Alabama to Virginia. The N–S trending suite, which crosscuts the NW trending suite, is unique to the southern Appalachians. Interpretation of the linear N–S oriented aeromagnetic anomalies beneath the Coastal Plain of South Carolina is inferred to be dikes that extend from near Charleston and into the Piedmont of North Carolina and Virginia.

Continuous exposures are rare in the Piedmont and only one site of crosscutting relationships has been observed. The field relationships of this site, although tenuous, suggest that the N–S trending dike, which crosscuts a NW trending dike, is younger (Ragland et al., 1983). The crosscutting relationships and different trends of the dikes suggest that they were possibly emplaced in two different stress regimes of differing ages.

A paleomagnetic study was undertaken to establish whether the time differential in the emplacement of the dike suites was significant or not and to determine relative ages. The dikes were carefully examined to see if systematic differences in their magnetic behaviour could be observed for this could have implications for their geological history.

Geology

The Piedmont province as well as the Triassic basins of central North Carolina contain numerous occurrences of diabase dikes (Fig. 1). Burt et al. (1978) have published a map of diabase locations in North Carolina which consists of a compilation of the known localities of diabase dikes as determined from field mapping and those interpreted from the USGS open file aeromagnetic data. In addition, the extent and occurrence of many more diabase dikes have also been recognized through the interpretation of aeromagnetic data elsewhere within the southern Appalachian Piedmont (Popenoe and Zietz, 1977; Bell et al., 1978, 1980; Daniels and Zietz, 1978; Daniels et al., 1983). Bell et al. (1978, 1980) in an aeromagnetic study of northeast South Carolina have demonstrated that NW trending aeromagnetic anomalies are caused by a group of closely spaced narrow dikes of varying dimensions. Popenoe and Zietz (1977) in a study of the geophysical basement beneath the Coastal Plain of South Carolina have interpreted numerous positive, narrow linear magnetic anomalies as being produced by two trends of diabase dikes which lie beneath the Coastal Plain and extend into the Piedmont province of South and North Carolina where they crosscut one another. One trend apparently is oriented NW which is the characteristic orientation of the exposed diabase dikes commonly found throughout the southern Appalachian Piedmont. The other trend, which is unique to the southeast U.S., has a radial fan shaped N–S orientation which appears to converge beneath the Coastal Plain towards the vicinity of Charleston

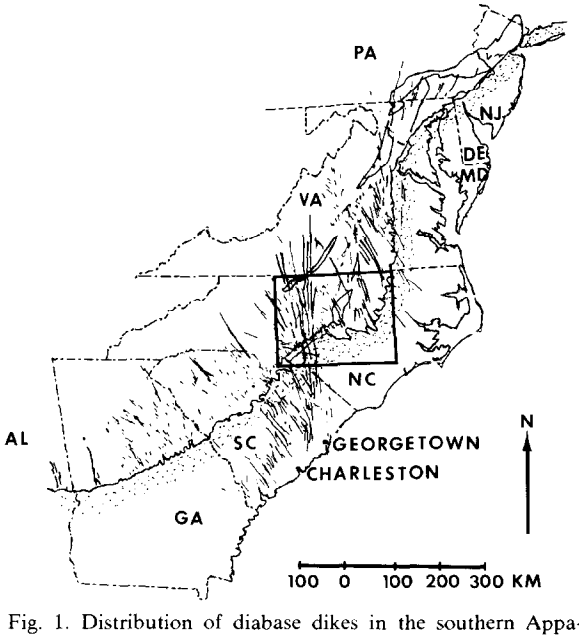


Fig. 1. Distribution of diabase dikes in the southern Appalachians (modified after Ragland et al., 1983). Triassic basins are outlined. Stipple pattern demarcates the extent of the Atlantic Coastal Plain. Area of study (see also Fig. 2) is outlined by the box.

and Georgetown, South Carolina (Fig. 1). Linear positive magnetic anomalies suggest that the N–S trend extends across the Piedmont of North Carolina and into Virginia (Daniels and Zietz, 1978; Daniels et al., 1983).

In the Wadesboro, Sanford, Deep River and Durham Triassic basins the NW trending dikes are more common whereas in the Dan River basin to the NW the N–S suite of dikes predominate. Within the Wadesboro basin, several of the dikes are possibly offset by intrabasin faults (Bain and Harvey, 1977). Dikes have also intruded into the Blue Ridge terrane of North Carolina and Virginia (Bryant and Reed, 1970) and have cut across the Brevard Zone in northeast Georgia (Lester and Allen, 1950).

The diabase dikes are commonly steeply dipping, unmetamorphosed and olivine normative tholeiites. The NW trending suite has an average trend in the range from N15° W to N42° W whereas the trend of the N–S suite typically ranges from N15° W to N10° E. In outcrop the black to grey diabase dikes typically have a reddish orange weathering rind and are often found as trains of spheroidally weathered boulders in roadcuts and streams.

The geochemistry of diabase dikes from eastern North America, especially North Carolina, has been the subject of several studies in order to determine correlations, if any, between the tectonic environment and chemistry (Ragland et al., 1968, 1971, 1983; Weigand, 1970; Weigand and Ragland, 1970, De Boer and Snider, 1979; Ragland and Whittington, 1983). Weigand and Ragland (1970) have subdivided the diabase dikes in eastern North America into two principal types: olivine normative tholeiite and quartz normative tholeiite. These can be further subdivided into four types on the basis of TiO_2 and large ion lithophile (LIL) content: a low and high LIL olivine tholeiite and a low and high TiO_2 quartz tholeiite (Ragland and Whittington, 1983) each having a distinct geographic distribution.

Ragland et al. (1983) have suggested that in North Carolina both the N–S and NW suites are olivine normative tholeiite with the N–S suite being slightly more enriched in the LIL elements and alkalies. They further point out that the N–S suite of diabase dikes are chemically similar to diabase in Virginia while the NW suite of dikes are similar to those NW trending diabase dikes in South Carolina.

Weigand and Ragland (1970) also pointed out that the olivine normative tholeiitic magma could be parental to the quartz normative tholeiitic magma (see also, Weigand and Ragland, 1970; Deininger et al., 1975; Smith et al., 1975; Ragland and Whittington, 1983; Ragland et al., 1983). Geochemical studies of subsurface basalt flows from the Clubhouse Crossroads boreholes near Charleston, S.C., however, show layers of olivine normative tholeiitic basalt flows intercalated within a sequence of quartz normative tholeiitic basalt flows (Gottfried et al., 1983). They concluded that this represents penecontemporaneous repetitive cycles of magma derivation due to varying degrees of melting in different thermal regimes, not necessarily a change in the tectonic environment.

De Boer and Snider (1979) considered the Carolinas to be the site of a Triassic–Jurassic hotspot on the basis of anomalous rare earth elements (La/Sm), high total iron and low TiO_2 content of diabase; however, there is no evidence of a plume track seaward.

Sampling and analytical techniques

A total of 204 cores were sampled from 29 diabase dike outcrops from the Piedmont of North Carolina. The sites are extensive and are from a majority of the Piedmont geological provinces and Triassic basins of North Carolina (Fig. 2). At least six to eight samples were obtained from each site with a portable water cooled, field drill with diamond impregnated bits. The cores were sampled as far apart as the outcrop allowed, usually of the order of 1 m or more. An aluminum orienting device and azimuthal compass were used to orient the cores at each site. In the laboratory each core was sliced with a diamond saw into one or more 22.2 ($\pm$0.5) mm length cylinders. The declination, inclination, and intensity of each specimen were then measured with a Digico complete results magnetometer interfaced with a DIGITAL MINC II minicomputer.

After NRM direction measurements, a pilot

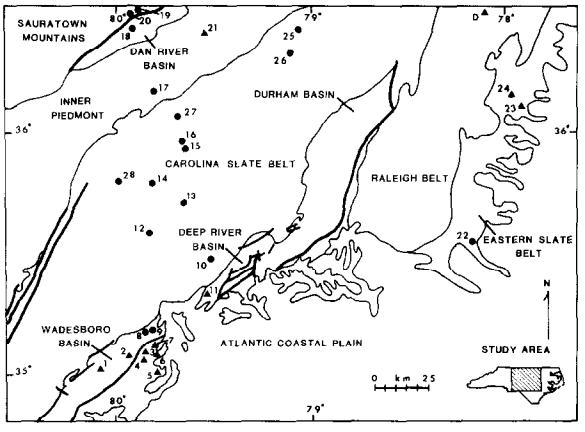


Fig. 2. Distribution of sampling sites of diabase dikes from the North Carolina Piedmont. The circles represent sites from N–S trending dikes and the triangles represent sites from NW trending dikes. Faults are shown as heavy lines.

specimen from each site was progressively demagnetized in alternating fields from 5 to 50 mT in 5 mT increments and also at 75 and 100 mT values. In addition one pilot from several sites was progressively thermally demagnetized from 100° to 500°C in 50°C increments and at 525° and 550°C.

The stage at which the characteristic magnetization was isolated was determined from the behavior of the demagnetization spectra and the stable end point curves of each site. This characteristic magnetization usually appeared after demagnetization in fields between 20 and 35 mT although at several sites the characteristic magnetization was recorded as low as 15 mT or as high as 45 mT and 500°C.

Discussion

AF demagnetization

The AF demagnetization spectra for several representative samples from the NW and N–S suites are shown in Fig. 3. The NRM intensities were in the range from 7.6×10^{-3} to 1.9×10^{-5} A/m averaging about 2.0×10^{-3} A/m. In a comparison of the coercivity spectrum of the two suites, approximately 20% of the average NRM intensity is removed by 15 mT for the N–S suite whereas for the NW suite 20% has been removed at earlier stages (< 5 mT) of AF demagnetization. The characteristic magnetization appears when 70% of the NRM intensity has been removed between 15 and 40 mT for the NW dikes and 60% of the NRM removed between 25 and 45 mT for the N–S dikes. The coercivity spectra are very typical of the other Mesozoic tholeiitic diabase intrusives located in eastern North America (De Boer, 1968; Beck, 1972; Watts, 1975; Smith, 1976; Volk, 1977; De Boer and Snider, 1979; Smith and Noltimier, 1979; Dooley and Smith, 1982).

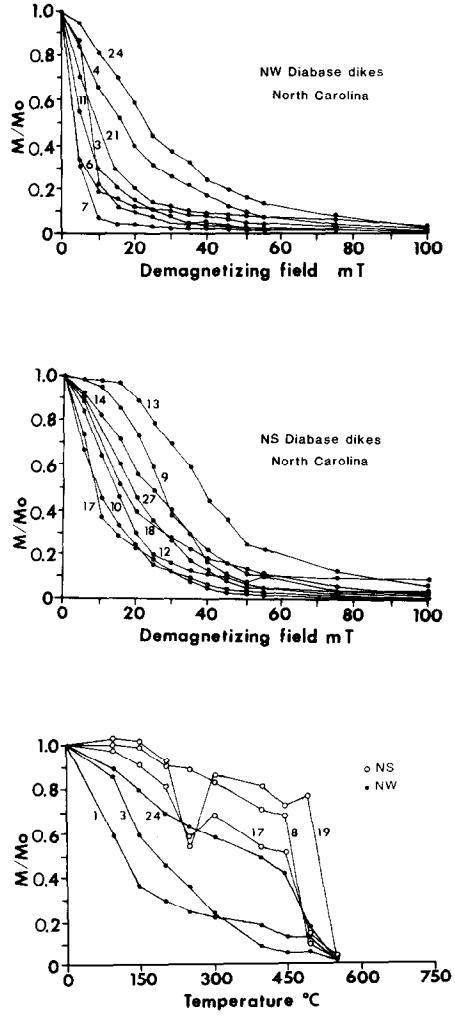


Fig. 3. Alternating field (mT) and thermal (°C) demagnetization spectra of representative samples for the NW suite and N–S suite.

Thermal demagnetization

Three pilot specimens from each suite were thermally demagnetized in order to compare the directions obtained with those from the AF demagnetization. The thermal demagnetization spectra for these sites are also shown in Fig. 3. The characteristic magnetization directions have been isolated for both suites in the range of 350° to 500°C.

The Curie point is in the range of 500 to 550°C indicative of titanomagnetite. There is a suggestion of an initial blocking temperature around 150° to 200°C in some but not all of the dikes.

Stable endpoint curves

Figure 4 shows stable endpoint curves (SEC) with normalized intensity for representative sites of the N–S (site 8) and NW (site 3) dikes. The AF and thermal demagnetization SEC for both sites shows a small soft component, probably a viscous magnetic component, being removed initially followed by a linear decrease towards zero as stable directions are attained with successive demagnetization.

A few of the thermally demagnetized sites exhibited an isolation of the stable remanence between 200° and 450°C followed by erratic or random directions with progressively higher demagnetization and no surviving thermally stable remanence. Stable directions for a few of the dikes were not isolated while some sites exhibited possible multicomponent behavior during demagnetization (Fig. 5).

A histogram showing the number of sites plotted versus the peak demagnetization (AF) at which the sites were considered stable is shown in Fig. 6. Unlike the dikes from South Carolina (Dooley and Smith, 1982) where a single maximum at 20 to 25 mT is observed the peak demagnetization for both the NW and N–S dike suites was bimodal at 20 to 25 mT and another maximum at 35 mT (see also De Boer, 1967).

Entire suites of diabase samples in many of the earlier studies in eastern North America were de-

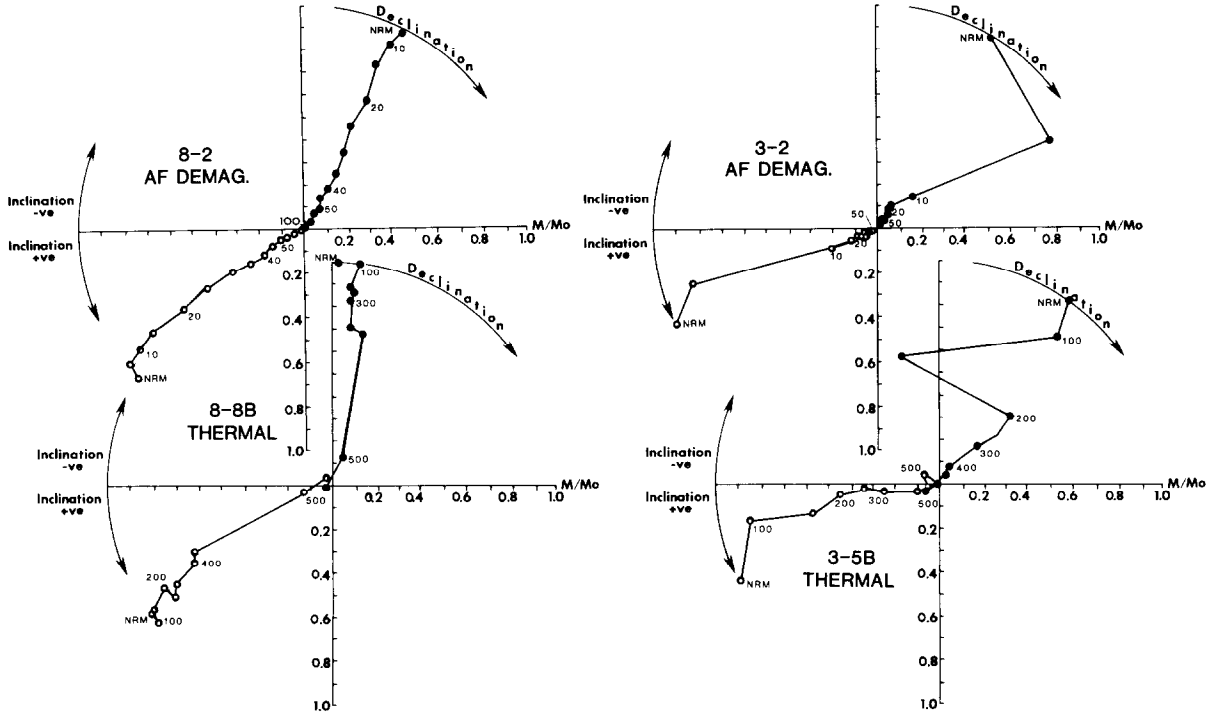


Fig. 4. Stable end point curves for a N–S dike (site 8) and a NW dike (site 3) for AF (mT, top) and thermal demagnetization (°C, bottom).

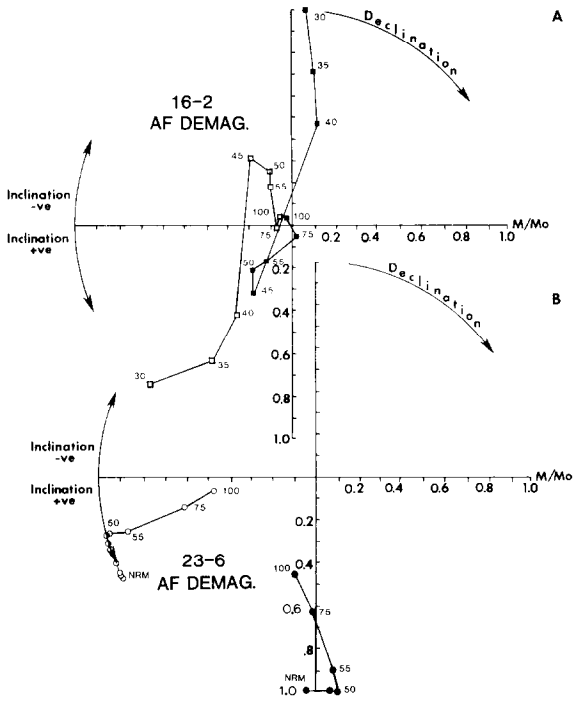


Fig. 5. Stable end point curves for AF demagnetization exhibiting possible multicomponent behaviour for a N–S dike (site 16) and a NW dike (site 23).

termined to have been magnetically clean or stable after batch AF demagnetization in a single peak field which ranged from 5 to 35 mT (Opdyke, 1961, Beck, 1965, 1972; Larochelle and Wanless, 1966; Larochelle, 1967, 1971; Carmichael and Palmer, 1968; Rigotti and Schmidt, 1976; Smith, 1976; Bell et al., 1980; Hodych and Hayatsu, 1980). The implications of this are that many of the directions and statistics obtained were possibly not the most magnetically stable or accurate and therefore contribute further to the error and problems of determining a comparable and reliable pole position for these rocks.

Several sites initially appear to have stable directions at 15 mT. However, with further AF cleaning, the directions change by only a few degrees and eventually stabilize at the higher peak demagnetization, which varied from 25 to 40 mT. Between 15 mT and the peak demagnetization there was not enough directional variation to warrant separation as distinct pole positions as they are within the error limits of the α_{95}. Therefore the best or stable direction was determined by

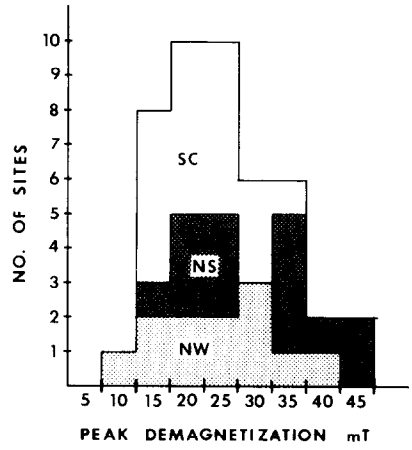


Fig. 6. Histogram of peak demagnetization values (mT) for dikes from the NW and N–S suites of North Carolina and dikes from South Carolina.

using the smallest circles of confidence (α_{95}) and the highest precision estimate (K) of Fisher (1953).

Stable directions

The NRM and cleaned directions are plotted for representative samples for both suites in Fig. 7. The site mean directions are shown in Fig. 8 and the statistical parameters are listed in Table 1. Also included in Fig. 8 is the site mean direction (D) for a NW diabase dike from the Raleigh belt of North Carolina (Delorey, 1983). Initially sites that had not achieved stable directions through the optimal cleaning stages either with AF or thermal demagnetization, therefore exhibiting highly disperse directions, were rejected. This led to the rejection of sites 25 and 26.

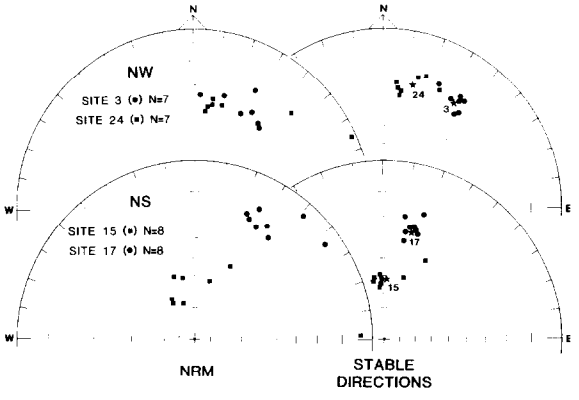


Fig. 7. NRM directions contrasted with the stable directions for representative sites from the NW and N–S suite.

When the directions with large α_{95} from this study were selectively included into the computation of the mean directions and pole positions there were no significant changes in the statistics and mean directions. Therefore, the calculation of the mean directions includes those sites with α_{95} less than 17°, which was the largest α_{95} value. The choice of α_{95} values used for rejection criteria from other studies of eastern North America diabase ranges considerably from 10° to 25° (Irving, 1964; Beck, 1972; Smith and Noltimier, 1979; Dooley and Smith, 1982). The effect of secular variation has probably been averaged out due to sampling of both suites from an extensive area of the North Carolina Piedmont (Fig. 2).

The results of site 23 were not included in the calculations of the NW mean directions as the stable site mean directions differ significantly from the NW mean directions (Table 1). The mean directions of this site are very similar to those directions determined for Carboniferous gabbros and diorites from South Carolina and North Carolina (Barton and Brown, 1983; Dooley, 1983). Therefore the possibility that site 23 is a gabbroic dike or plug of carboniferous age rather than a Mesozoic age dike must be considered.

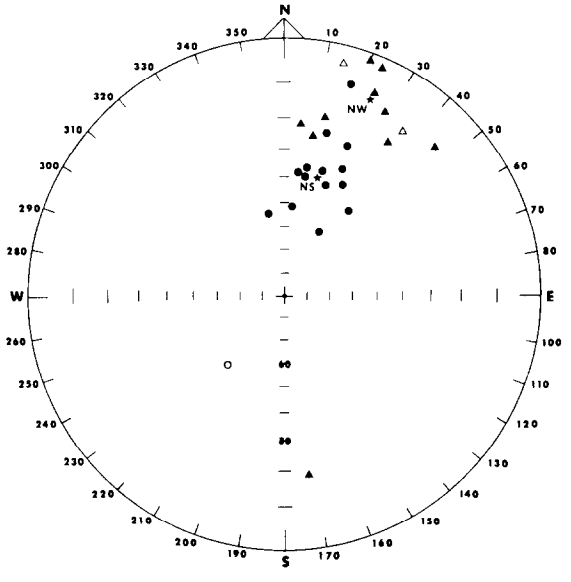


Fig. 8. Mean directions for 12 NW sites (triangles) and 15 N–S sites (circles). The open symbols are negative inclinations. The star represents the respective means for each suite (see Table 1).

Site 22 ($D = 218.1°$, $I = -51.2°$, $\alpha_{95} = 9.3°$) is a dike that was apparently magnetized within a reverse field. Dikes with reverse directions have been reported in a few studies of eastern North America diabase (Choiniere and Watkins, 1976; Smith, 1977; Smith and Noltimier, 1979). The directions, although reversed, were included in the calculation of the mean directions of the N–S suite of dikes.

Grouped together the individual site means from both suites form an elongate grouping suggesting that the mean directions are streaked towards the direction of the present day axial geocentric dipole field and therefore not thoroughly cleaned. However, considered as two distinctive dike trends, two distinct groups can be delineated. The interangle between the two site mean directions of these two groups is approximately 30°, which is greater than the sum of the respective α_{95} values.

The principal difference between the mean direction of the NW dikes ($D = 23.1°$, $I = 9.6°$, $\alpha_{95} = 9.7°$) and the N–S dikes ($D = 17.7°$, $I = 39.1°$, $\alpha_{95} = 7.7°$) is that the magnetic inclination of the former is much shallower. This eventually translates into a higher latitude and therefore possibly younger pole position for the N–S dike suite.

Pole positions

As mentioned above the mean pole positions calculated from averaging the VGP's of the NW and N–S diabas suites are statistically different (Table 2 and Fig. 9). These pole positions are listed in Table 3 for comparison with other pole positions obtained for Late Triassic–Early Jurassic age igneous rocks from the Appalachian Orogen.

The overall pole position of Mesozoic diabase from the southern Piedmont (Watts, 1975; Bell et al., 1980; Dooley and Smith, 1982) is 66.0° N, 76.1° E ($N = 76$, $K = 28$, $A_{95} = 3.1°$) and lies within the limits of Early Jurassic poles of the APW path (Fig. 9).

Also shown in Fig. 9 is an apparent polar wander (APW) path for North America using poles from the stable craton of North America

TABLE 1

Summary of site mean directions and statistical parameters for the NW and N–S diabase suites of North Carolina

Site	n/N	D	I	α_{95}	K	R	VGP		Demag.	Site	
							Lat.° N	Long.° E		Lat.° N	Long.° W
Northwest dikes											
1	5/6	29.2	11.8	8.4	85	4.95	50.4	50.2	20 mT	35.0	80.1
2	6/6	10.1	25.7	6.6	106	5.95	66.6	74.7	30 mT	35.1	79.9
3	7/7	33.7	18.4	3.6	278	6.98	50.0	41.8	25 mT	35.1	79.9
4	8/8	20.0	1.6	6.6	71	7.90	51.0	67.2	30 mT	35.1	79.9
5	7/7	35.7	−12.6	10.5	34	6.83	36.7	53.9	15 mT	35.0	79.8
6	8/8	46.0	10.7	7.3	59	7.88	38.3	34.2	40 mT	35.0	79.8
7	7/8	13.4	−3.9	11.2	30	6.80	50.9	78.7	25 mT	35.1	79.8
11	8/8	23.2	2.1	6.8	68	7.90	49.5	63.1	20 mT	35.3	79.6
21	6/7	5.6	22.4	9.1	55	5.91	64.7	87.6	30 mT	36.4	79.5
*23	4/6	172.0	17.9	6.0	238	3.99	44.3	113.1	25 mT	36.0	77.9
24	7/7	12.5	19.1	5.9	107	6.94	61.4	75.6	35 mT	36.0	77.9
D	8/9	24.3	8.0	4.8	134	–	50.6	61.8	15 mT	36.5	77.9
North–south dikes											
8	8/8	25.5	33.2	3.8	209	7.97	61.7	40.6	35 mT	35.2	79.8
9	6/6	16.7	8.4	3.6	343	5.99	55.4	69.9	20 mT	35.2	79.8
10	7/7	16.5	36.4	13.0	23	6.73	69.0	52.5	25 mT	35.5	79.5
12	8/8	6.9	39.3	3.6	244	7.97	75.3	74.2	20 mT	35.7	79.9
13	7/7	20.7	41.3	5.1	144	6.96	68.4	38.7	40 mT	35.8	79.7
14	6/8	21.5	25.1	9.6	50	5.90	60.2	54.4	35 mT	35.9	79.7
15	8/8	3.7	51.2	7.6	54	7.87	84.8	62.9	25 mT	36.0	79.6
16	7/7	28.1	58.8	16.2	15	6.59	67.6	172.8	35 mT	36.0	79.6
17	8/8	14.6	23.4	3.7	223	7.97	62.6	67.8	35 mT	36.2	79.8
18	6/6	36.7	43.5	13.6	25	5.80	56.9	19.1	45 mT	36.4	79.9
19	5/5	9.4	37.5	6.2	154	4.97	69.7	72.5	45 mT	36.5	79.9
20	6/7	27.9	36.7	9.4	52	5.90	60.9	35.9	20 mT	36.5	79.9
22	6/6	218.1	−51.2	9.3	53	5.91	58.3	7.1	500° C	35.6	78.5
*25	6/6	–	–	–	–	–	–	–	–	36.4	79.0
*26	8/8	–	–	–	–	–	–	–	–	36.3	79.1
27	6/6	347.5	53.7	4.1	269	5.98	79.6	183.2	15 mT	36.2	79.6
28	6/6	9.6	37.7	5.4	156	5.97	73.0	68.1	25 mT	36.0	79.9

Site	n/N	D	I	α_{95}	K	R	Pole position		dp	dm
							Lat.° N	Long.° E		
Means										
NW	11/12	23.1	9.6	9.7	23	10.57	52.6	60.7	4.9	9.8
N–S	15/17	17.7	39.1	7.7	26	14.46	69.6	47.1	5.5	9.2
All	26/29	20.4	26.8	7.9	14	24.20	61.9	55.3	4.7	8.6

n/N is the number of samples used in the computation of the mean (n) and the total number of cores tested (N). D and I are the mean declination and inclination of the site respectively. α_{95}, K and R are statistical parameters of Fisher (1953).

* indicates sites that are not included in the computation of the mean.

(Table 4). This APW path shows two phases of consistent apparent polar motion representing (1) westward, Early Triassic–Late Triassic and (2) eastward, Early Jurassic–Middle Jurassic displacement of the pole of North America. The bend in the APW path (poles *C*, *K*, and *G*1, Fig. 9) represents the change from the eastward to westward displacement of the North American pole. This also corresponds to the rotation of North America against the relatively stationary African

plate during the incipient opening of the Atlantic Ocean (Steiner, 1975, 1983).

A significantly different APW path, based on paleomagnetic Euler poles (Gordon et al., 1984), was also considered. However, the reliability of one of the pole positions used by them has been questioned (Steiner, 1983, p. 3). Therefore, the APW path of Gordon et al. (1984) was not used in this study.

Several radiometric studies of Mesozoic igneous rocks of eastern North America provide ages between 175 to 200 m.y. with clustering at 175 to 180 m.y. and 195 to 200 m.y. (Erickson and Kulp, 1961; Carmichael and Palmer, 1968; De Boer, 1968; Armstrong and Besancon, 1970; Wanless and Stevens, 1971; Wanless et al., 1972; Watts and Noltimier, 1974; Dallmeyer, 1975; Deininger et al., 1975; Watts, 1975; Dooley, 1977; Hayatsu, 1979; Sutter and Smith, 1979; Dooley and Smith, 1982; Dooley and Wampler, 1983; Lanphere, 1983). Deininger et al. (1975) have reported ages of 161 to 168 m.y. for diabase from Alabama. K–Ar age determinations of diabase dikes from South Carolina (Dooley and Smith, 1982) suggest a predominant age grouping at 194 m.y. However, there is also a suggestion of a 180 m.y. age group. In addition several conventional K–Ar studies report ages that are greater than 200 m.y. (e.g., Carmichael and Palmer, 1968; Armstrong and Besancon, 1970; Dooley and Smith, 1982) but there have been no $^{40}Ar/^{39}Ar$ determinations which would establish this age grouping.

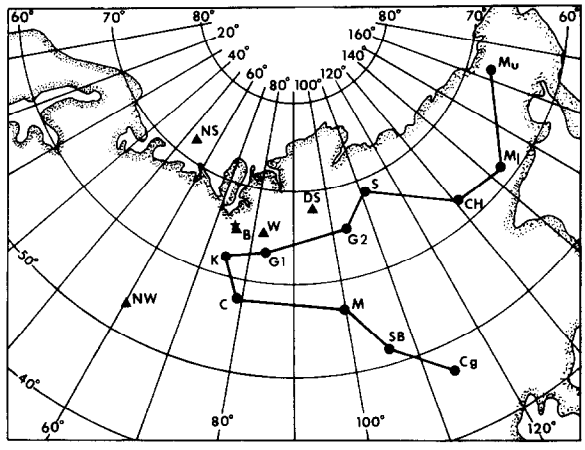


Fig. 9. Pole positions of Early Mesozoic diabase dikes from the southern Appalachian Piedmont (triangles; Table 2) plotted relative to the Early Mesozoic apparent polar wander path for North America (solid circles; Table 3). The star represents the mean pole position for diabase from the southeastern U.S. (*Cg* —Chugwater, *SB*—State Bridge, *M*—Moenkopi, *C*—Chinle, *K*—Kayenta, *G1*—Group 1, *G2*—Group 2, *S*—Summerville, *CH*—Canelo Hills, *Ml*—Lower Morrison, *Mu*—Upper Morrison, *W*—Watts (1975), *B*—Bell et al. (1980), *DS*—Dooley and Smith (1982), *NW*—northwest suite, and *NS*—north–south suite).

It is difficult to explain the aberrant displacement of the NW and N–S pole positions from the Early Jurassic APW path. To bring these points into relatively near coincidence with the APW path would require a tilting of the North Carolina Piedmont westwards about 10° to 15° such as that proposed by Dooley and Smith (1982). Based on a paleomagnetic study of diabase from South

TABLE 2

Mean pole positions for southeastern North American Early Mesozoic igneous rocks

N	*K*	A_{95} *	Lat.°N	Long.°E	Source
25	126	2.6	65.1	82.3	Watts (1975) [a]
5	78	8.7	64.5	75.7	Bell et al. (1980) [b]
20	24	6.8	67.9	95.3	Dooley and Smith (1982) [c]
11	35	7.8	52.8	60.2	NW dikes, this study [d]
15	23	8.1	71.5	53.5	N–S dikes, this study [d]
76	28	3.1	66.0	76.1	All southeast U.S.

* Pole position for each study was determined from the VGP's of each site and is therefore the A_{95}.

[a] The pole position of Watts (1975) is from diabase sites from Georgia, South Carolina and North Carolina.

[b] Pole position of Bell et al. (1980) is from diabase sites in the Haile–Brewer area of South Carolina.

[c] Pole position of Dooley and Smith (1982) based on 20 diabase sites collected in the South Carolina Piedmont. (The individual site VGP's were recalculated due to an error in the initial calculations.)

[d] Pole position of this study based on 11 NW dikes and 15 N–S dikes.

TABLE 3

Pole positions of northern, central and southern Appalachian Mesozoic igneous rocks

	Lat° N	Long° E	Confidence limits *	Source
Northern Appalachians				
Anticosti diabase dikes, Anticosti Island	75.7	84.7	1.2	Larochelle (1971)
Avalon dikes and sills, Newfoundland	72.9	87.8	3.0, 4.3	Hodych and Hayatsu (1980)
Shelburne dike, Nova Scotia	69.0	98.0	3.5, 5.5	Larochelle and Wanless (1966)
North Mountain Basalt, Nova Scotia	66.0	113.0	6.0, 10.5	Larochelle (1967)
North Mountain Basalt, Nova Scotia	73.0	104.0	4.5, 6.5	Carmichael and Palmer (1968)
Grand Manan Lavas, New Brunswick	80.5	100.5	17.0, 23.5	Carmichael and Palmer (1968)
Caraquet diabase dike (C), New Brunswick	78.0	143.0	9.0, 12.0	Sequin et al. (1981)
Caraquet diabase dike (N), New Brunswick	74.0	133.0	7.0, 11.0	Sequin et al. (1981)
Caraquet diabase dike (T), New Brunswick	62.0	82.0	5.0, 8.0	Sequin et al. (1981)
Central Appalachians				
Granby and Holyoke lavas, Massachusetts	55.0	88.0	6.0, 11.0	Irving and Banks (1961)
Diabase dikes and sills, Connecticut and Maryland	68.6	100.9	1.6	Smith (1976)
Group 1	63.0	83.2	2.3	Smith and Noltimier (1979)
Group 2	65.3	103.2	1.4	Smith and Noltimier (1979)
Connecticut Valley sills and dikes	64.2	86.4	8.8, 11.0	De Boer (1968)
Diabase, Pennsylvania	62.0	104.5	1.5	Beck (1972)
Diabase and basalt flows, New Jersey	63.6	98.5	4.3, 7.8	Opdyke (1961) [a]
Palisades Sill	74.4	97.5	–	Rigotti and Schmidt (1976)
Southern Appalachians				
Clubhouse Crossroads basalt flows, South Carolina	(inclination = 35.4 ± 6.4°)			Phillips (1983) [b]
Diabase dikes, Georgia and North Carolina	65.1	82.3	2.6	Watts (1975)
Haile–Brewer diabase dikes, South Carolina	64.5	75.7	8.7	Bell et al. (1980) [c]
Diabase dikes, South Carolina	67.9	95.3	6.8	Dooley and Smith (1982) [c]
Northwest diabase dikes, North Carolina	52.8	60.2	7.8	This study
North–south diabase dikes, North Carolina	71.5	53.5	8.1	This study

* Confidence limits are in degrees as either dp and dm or α_{95} (Northern and Central Appalachians) and A_{95} (Southern Appalachians).

[a] Recalculated for the lavas and intrusives.

[b] Phillips (1983) used unoriented cores from borehole cores. He obtained an inclination and colatitude which are similar to those obtained from the Mesozoic dikes in South and North Carolina.

[c] Recalculated for pole positions and A_{95}.

TABLE 4

North America Mesozoic pole positions

Unit location	Age	Lat° N	Long° E	Confidence *	Source
Chugwater Fm., Wyoming [a]	ETr	46.8	113.7	–	Shive et al. (1984)
State Bridge Fm., Colorado	ETr	51.6	105.7	1.9, 3.8	Christensen (1974)
Moenkopi Fm., Wyoming [b]	ETr	56.7	100.0	–	Shive et al. (1984)
Chinle Fm., New Mexico	LTr	57.7	79.1	2.1, 4.2	Reeve and Helsley (1972)
Kayenta Fm., Utah	EJ	61.9	74.4	3.7, 7.1	Steiner and Helsley (1974)
Group 1, NE U.S.A.	EJ	63.0	83.2	2.3	Smith and Noltimier (1979)
Group 2, NE U.S.A.	EJ	65.3	103.2	1.4	Smith and Noltimier (1979)
Summerville Fm., Utah	MJ	67.5	110.6	3.2, 5.4	Steiner (1978)
Canelo Hills Volcanics, Arizona	LJ	62.2	130.3	4.1, 7.4	Kluth et al. (1982)
Lower Morrison Fm., Colorado	LJ	61.4	142.2	4.0, 6.5	Steiner and Helsley (1975)
Upper Morrison Fm., Colorado	LJ	67.5	161.8	3.5, 5.0	Steiner and Helsley (1975)

* Confidence limits are in degrees as either dp and dm or α_{95}.

[a] Chugwater mean is the average of VGP's from Shive et al. (1984); Herrero-Bervera and Helsley (1983); and Van der Voo and Grubbs (1977).

[b] Moenkopi mean is the average as determined by Shive et al. (1984), of data from Helsley (1969); Helsley and Steiner (1974); Baag and Helsley (1974); and Purucker et al. (1980).

Carolina, they considered a tilting model of the Piedmont and concluded that the tilting due to updoming, isostatic adjustment and differential erosion, each producing variable amounts of complex tilting, was not resolvable in the amount of tilting but that the sense of tilting was constrained to the west. The amount of tilting required to bring the NW and N–S pole positions into near coincidence with the APW path also displaces the other southern Appalachian diabase pole positions (Watts, 1975; Bell et al., 1980; Dooley and Smith, 1982) away from the APW path. There is no clear evidence of a quantitative amount of tilting of or between the North Carolina and South Carolina Piedmont in post-Late Jurassic.

The conclusion from the field relationships and this paleomagnetic study is that the N–S trending dikes are probably younger than the NW trending dikes. The pole positions of the NW and N–S suites are statistically different from one another with the NW suite having a pole position with a lower latitude (therefore possibly older) than the VGP latitude of the N–S suite when compared to the APW path of North America.

The tectonic effects of the rifting environment in the southeast U.S. are not very well understood with respect to the timing of the emplacement of the dike suites. Several authors (De Boer and Snider, 1979; Manspeizer, 1981; Swanson, 1982; Bedard, 1985) have proposed models of rifting of eastern North America which can significantly account for the distribution of the NW trending diabase dikes in the southeast U.S. However, the disposition of the N–S trending suite of diabase

dikes cannot be successfully modeled based upon these simplified models. Further resolution of the relative ages can better be resolved by reliable and necessary radiometric age dating of the two dike suites.

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